

RNA-Seq Analysis Pipeline

Original Authors' Pipeline

The original study used the following RNA-seq analysis pipeline:

RNA extraction → Library preparation → Poly(A) enrichment → Illumina sequencing → TopHat alignment → Gene read counting → DESeq → Differentially expressed genes → IPA pathway analysis.

The original study used an RNA-seq pipeline to analyze gene expression changes in mice infected with *Staphylococcus aureus*. Total RNA was extracted from mouse ear tissue, followed by RNA library preparation and poly(A) enrichment. The samples were then sequenced using the Illumina HiSeq platform to generate 100-bp paired-end reads. The sequencing reads were aligned to the mouse reference genome using TopHat, and gene read counts were generated from the alignments. Differential gene expression was analyzed using DESeq, and the resulting differentially expressed genes were further examined using Ingenuity Pathway Analysis (IPA) to identify affected biological pathways and regulatory processes.

Our Galaxy Re-analysis Pipeline

Our group used the following Galaxy workflow:

NCBI SRA → Raw paired-end RNA-seq reads → FastQC + MultiQC → Quality assessment → No trimming → Mus musculus reference genome → TopHat → BAM files → featureCounts → Gene-level count matrix → DESeq2 → Differentially expressed genes → PCA, MA plot, dispersion plot, and p-value histogram

Four RNA-seq samples were analyzed: two control samples and two infected samples.

FastQC and MultiQC showed less than 0.1% adapter contamination and no poor-quality reads. Therefore, trimming was not performed.

Our group used the *Mus musculus* reference genome for the analysis. Reads were aligned using **TopHat**, and the resulting BAM files were processed using **featureCounts** to obtain gene-level read counts. Differential expression analysis was then performed using **DESeq2**.